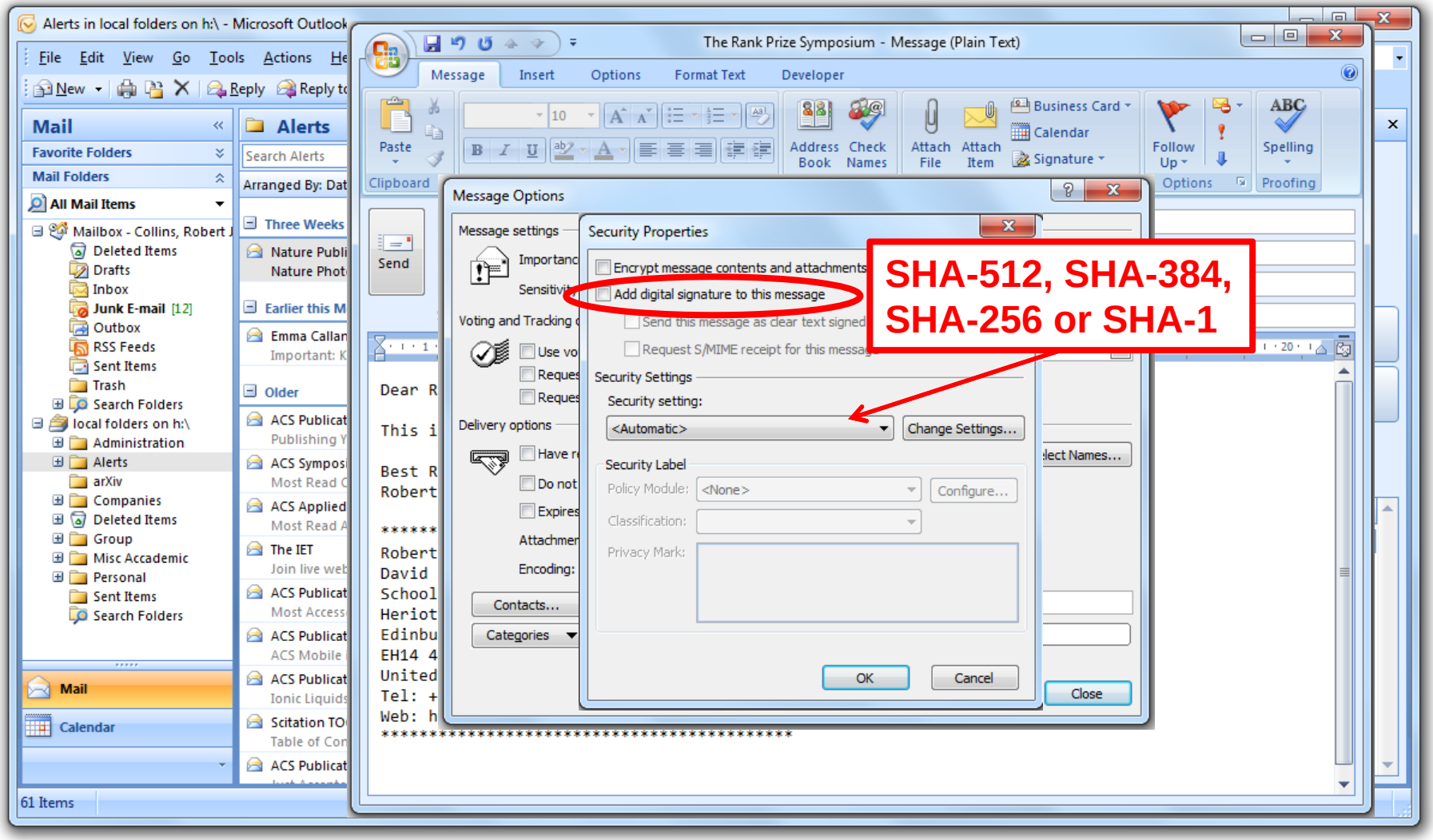
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University of Strathclyde Glasgow

Advances in Experimental Quantum Digital Signatures

Ross J. Donaldson[1], Robert J. Collins[1], Klaudia Kleczkowska[1], Ryan Amiri[1], Petros Wallden[1,2], Vedran Dunjko[3,4], Erika Andersson[1], John Jeffers[5] and Gerald S. Buller[1]

[1]Institute of Photonics & Quantum Sciences, Heriot-Watt University, Edinburgh, United Kingdom
[2]School of Informatics, The University of Edinburgh, Edinburgh, United Kingdom
[3]Institute for Quantum Optics & Quantum Information, Austrian Academy of Sciences, Innsbruck, Austria
[4]Division of Molecular Biology, Ruder Bošković Institute, Zagreb, Croatia
[5]Department of Physics, University of Strathclyde, Glasgow, United Kingdom

Digital signatures



“Standard” Protocols

- RSA
  (Rivest Shamir Alderman - IEEE 1363)
- DSA
  (Digital Signature Algorithm - FIPS PUB 186-*X*)
  - Originally included SHA-1
- ECDSA
  (Elliptic Curve Digital Signature Algorithm - ANSI X9.62)
- Security relies on computational assumptions
  - Can be retrospectively altered by future advances in algorithms or computational hardware

Quantum digital signatures

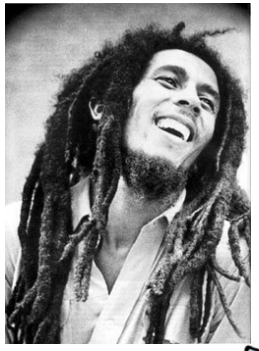
- Alice's (sender) "quantum signatures" are randomly selected sequences of quantum states

$$|\psi_{i1}^0\rangle \otimes |\psi_{i2}^0\rangle \otimes \cdots \otimes |\psi_{iM}^0\rangle \quad \text{for "0"}$$

$$|\psi_{i1}^1\rangle \otimes |\psi_{i2}^1\rangle \otimes \cdots \otimes |\psi_{iM}^1\rangle \quad \text{for "1"}$$

which are sent to Bob and Charlie (receivers).





- There are many states used to "sign" a single bit.
- Nobody except Alice knows exactly what these states are.
- Message transmission may take place a long time after the quantum states are sent.



- Information theoretically secure

PRA **74** 022304 (2006)

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Quantum Digital Signatures — 2001
Gottesman, & Chuang
arXiv: quant-ph/0105032

Experimentally realizable quantum comparison of coherent states and its applications — 2006
Erika Andersson, M. Curty, & I. Jex
Physical Review A

Experimental demonstration of quantum digital signatures using phase-encoded coherent states of light — 2012
PJ. Clarke, RJ. Collins, V. Dunjko, Erika Andersson, John Jeffers, & Gerald Buller
Nature Communications

Realisation of quantum digital signatures without the requirement of quantum memory — 2014
RJ. Collins, RJ. Donaldson, V. Dunjko, P. Wallden, PJ. Clarke, Erika Andersson, John Jeffers, & Gerald Buller
Physical Review Letters

Experimental demonstration of kilometer-range quantum digital signatures — 2015
RJ. Donaldson, RJ. Collins, K. Kleczkowska, R. Amiri, P. Wallden, V. Dunjko, John Jeffers, Erika Andersson, & Gerald Buller
In process of revision with Physical Review Letters
arXiv - 1509.07827[quant-ph]

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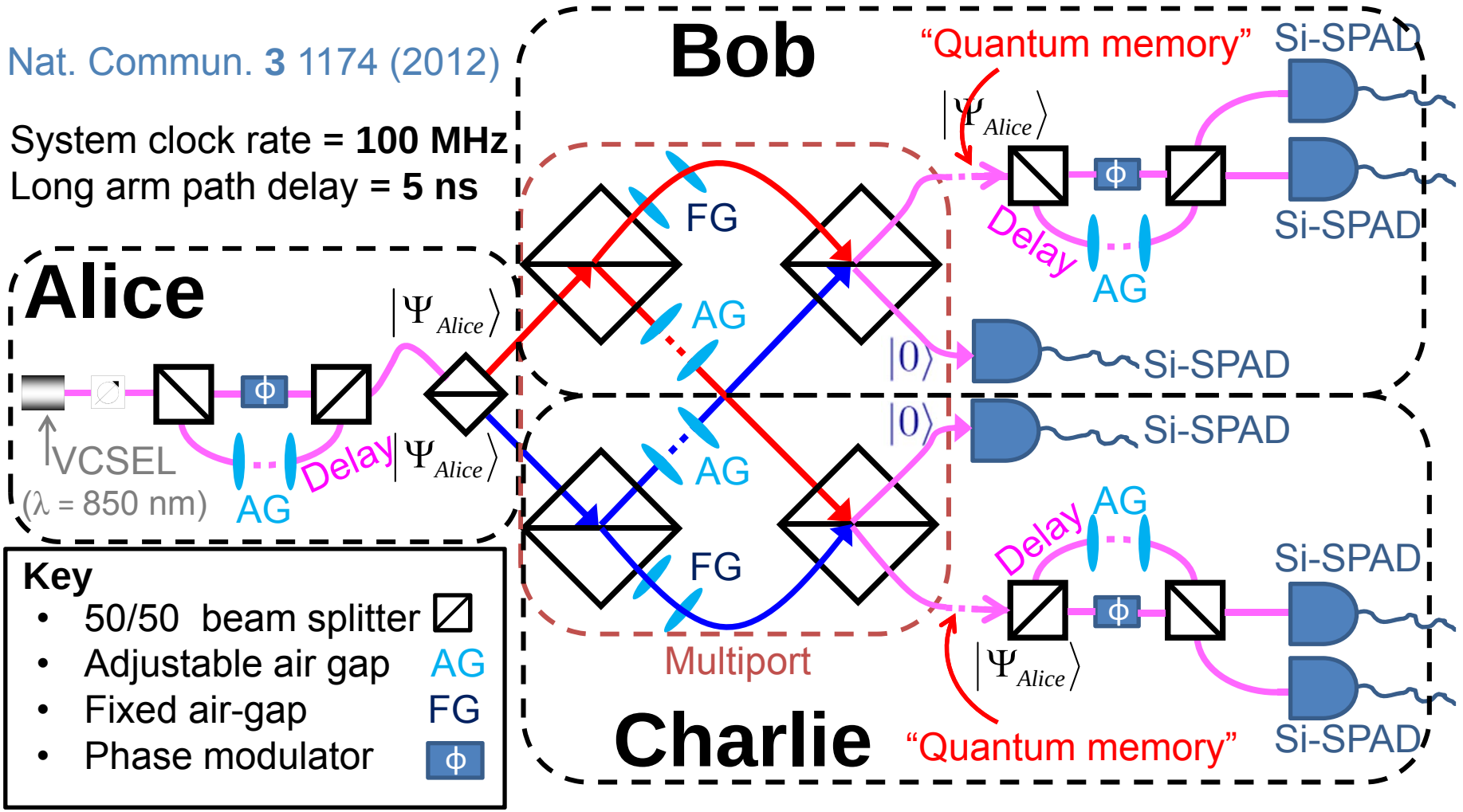
Outline of presentation

- Initial designs using active state discrimination
- Passive state discrimination/elimination
- Improved symmetrization protocols
- Future outlook for experiments

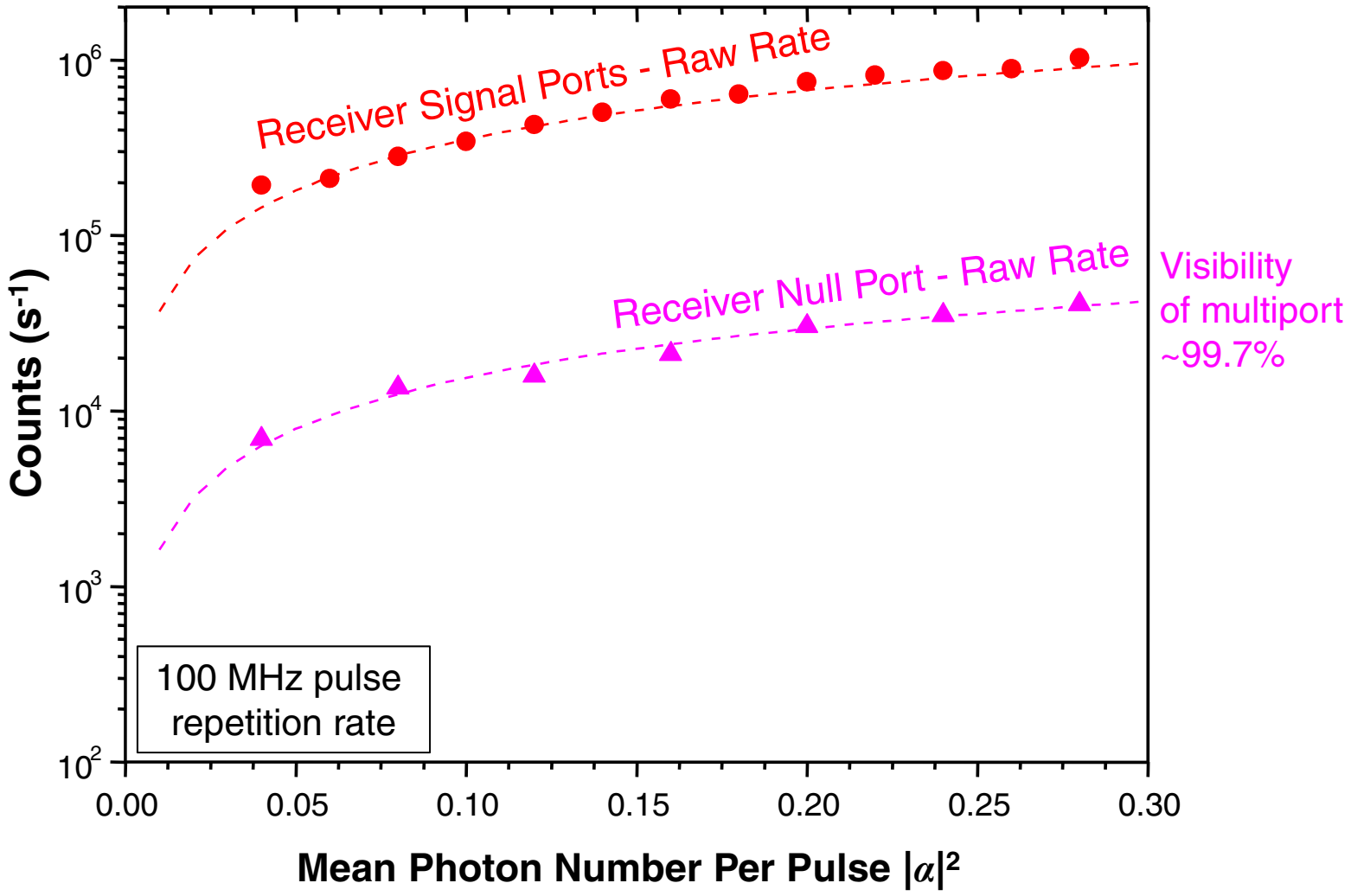
Quantum Digital Signatures: Active state discrimination

Nat. Commun. **3** 1174 (2012)

System clock rate = **100 MHz**
Long arm path delay = **5 ns**



QDS: Active state discrimination



Nat. Commun. **3** 1174 (2012)

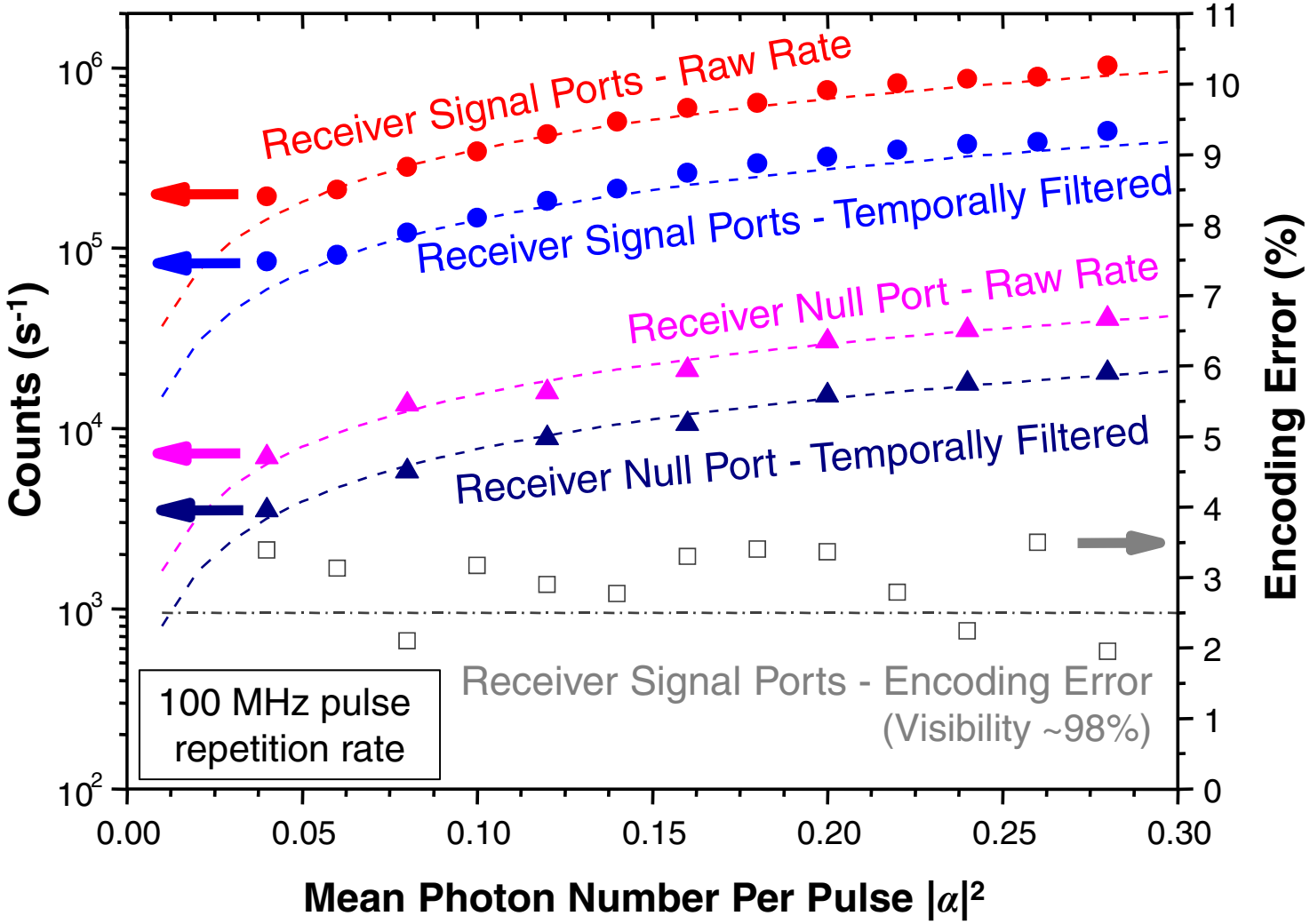
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QDS: Active state discrimination

Takes Alice 0.691 s to send a half bit



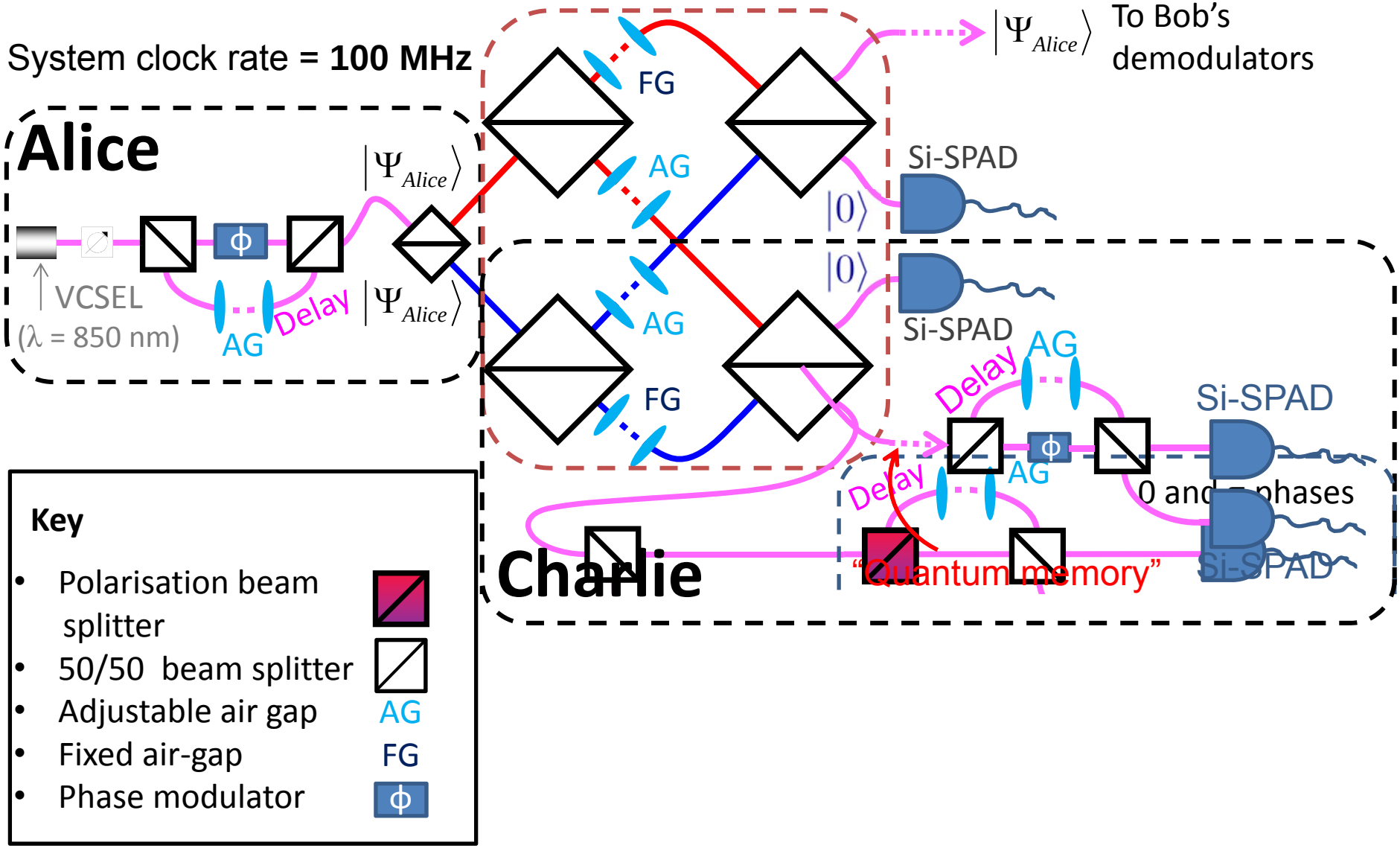
Nat. Commun. **3** 1174 (2012)

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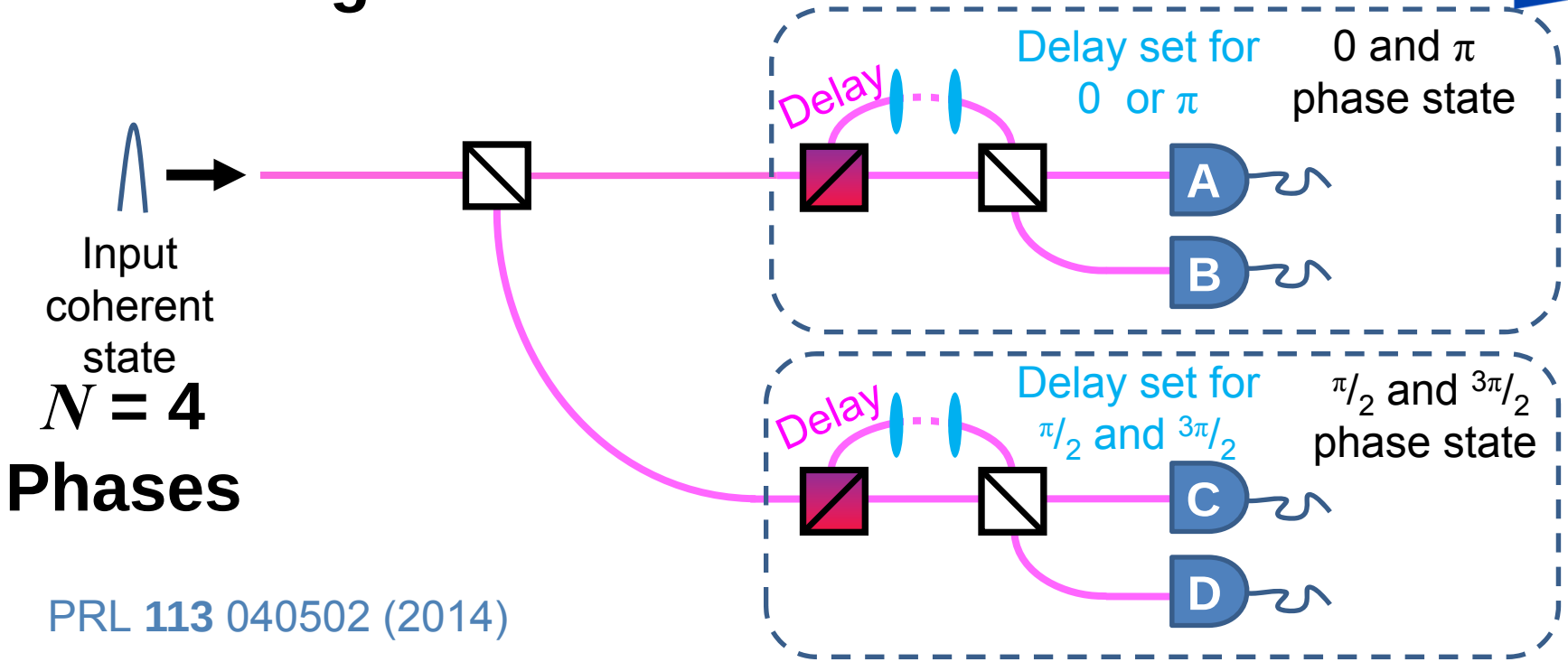
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Active state discrimination



PRL **113** 040502 (2014)

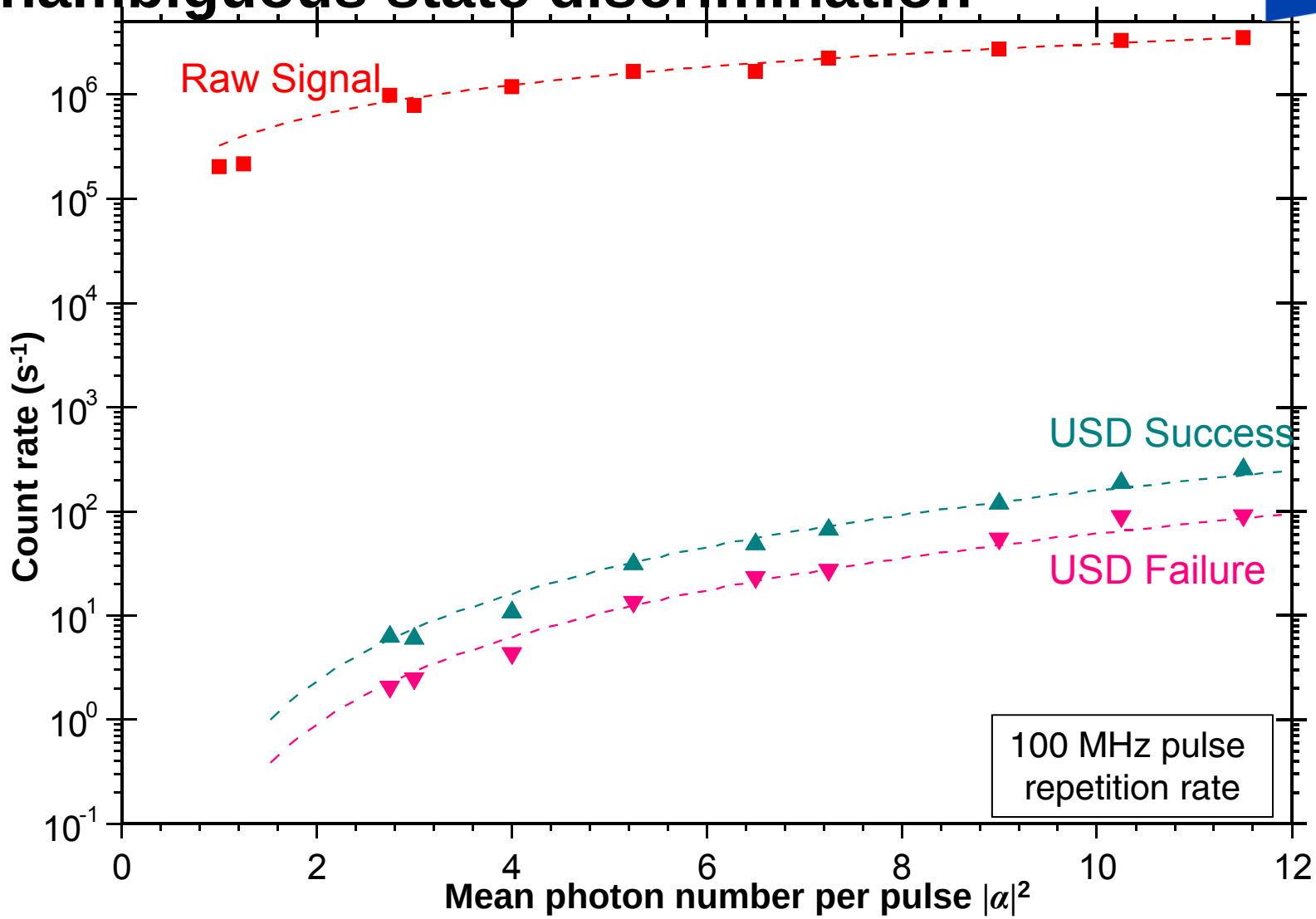
Unambiguous state elimination



PRL **113** 040502 (2014)

Incident Phase	Detector A ("Not π") Fires	Detector B ("Not 0") Fires	Detector C ("Not $^{3\pi}/_{2}$") Fires	Detector D ("Not $^{\pi}/_{2}$") Fires
0	Possible	Excluded	Possible	Possible
$^{\pi}/_{2}$	Possible	Possible	Possible	Excluded
π	Excluded	Possible	Possible	Possible
$^{3\pi}/_{2}$	Possible	Possible	Excluded	Possible

Unambiguous state discrimination



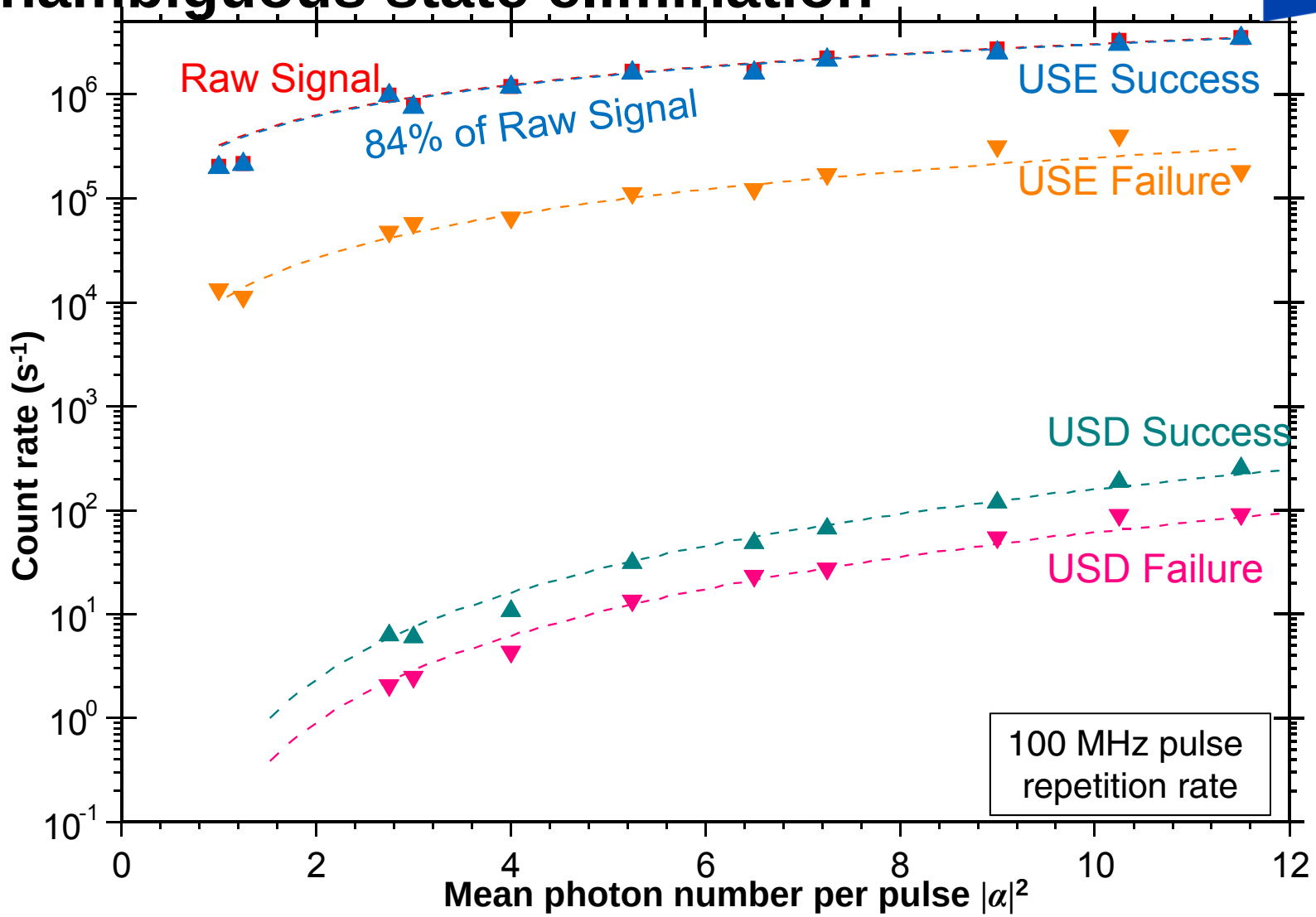
PRL **113** 040502 (2014)

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Unambiguous state elimination



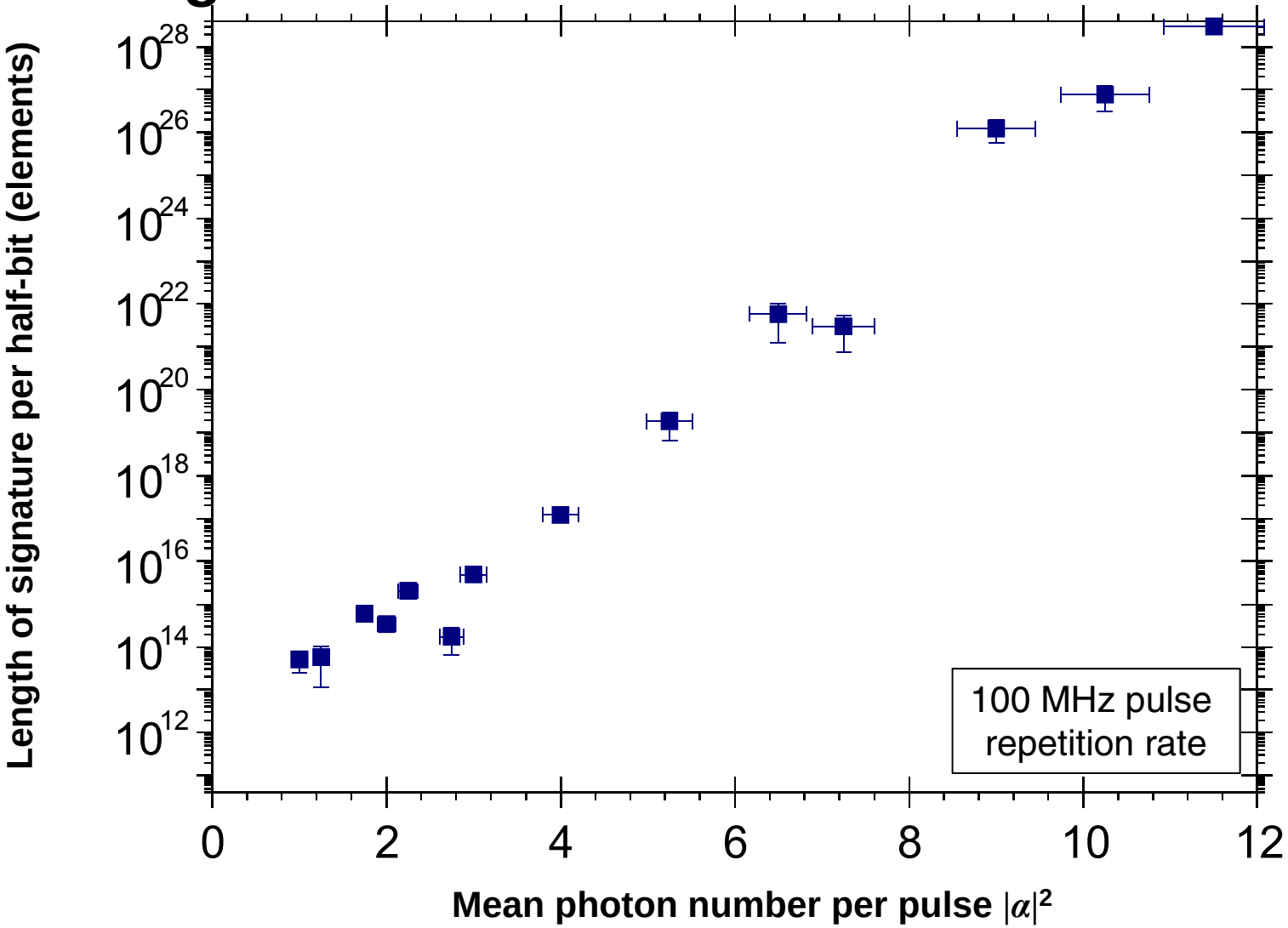
PRL **113** 040502 (2014)

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Unambiguous state elimination



PRL **113** 040502 (2014)

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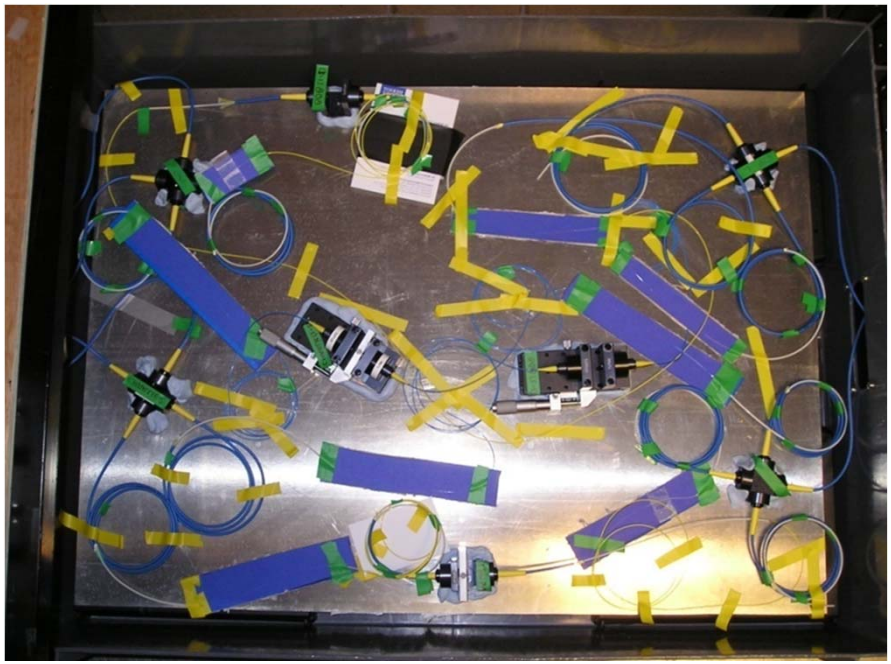
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The “problem” of the multiport

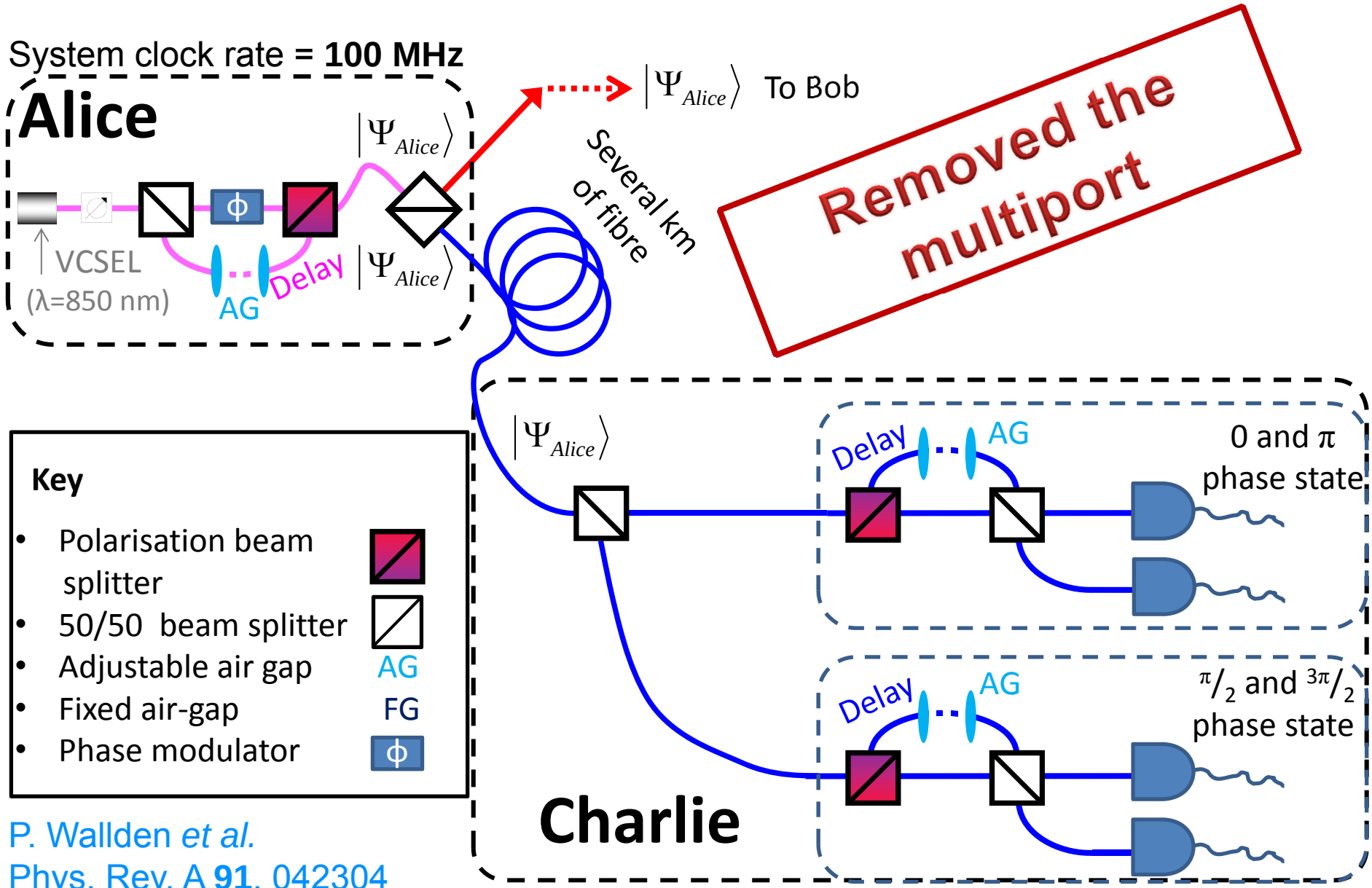
Symmetrization is carried out using a fiber based multiport

- The multiport has ~8 dB loss
- The relative path-lengths can drift
  - This results in misrouted pulses
  - The multiport is small to make stabilization easier
  - Alice, Bob and Charlie are separated by ~5 meters
- We require a revised protocol for long distance communications
  - P. Wallden *et al.* Phys. Rev. A **91**, 042304



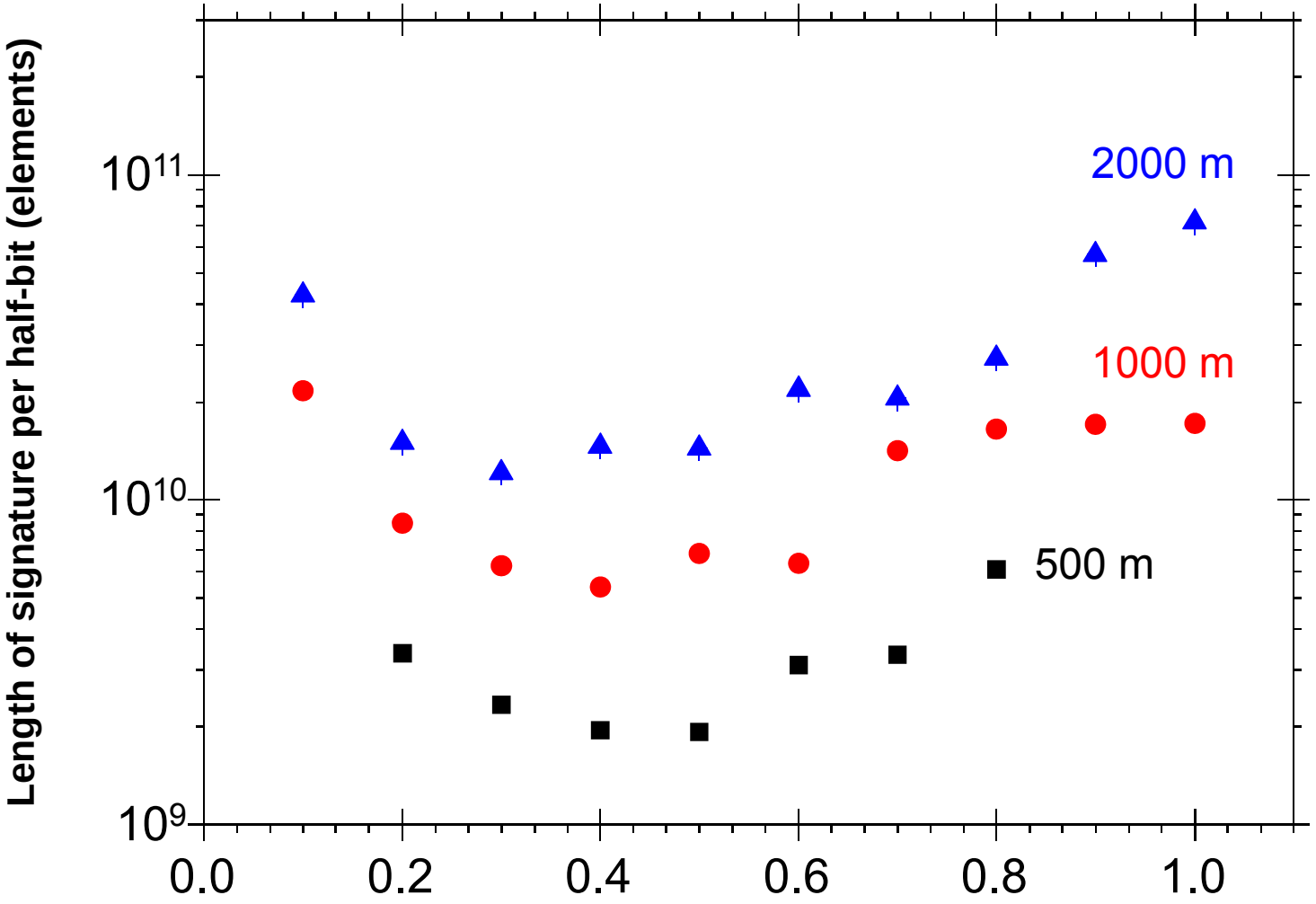
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Extended distance QDS using QKD technology



P. Wallden *et al.*
Phys. Rev. A **91**, 042304

Extended transmission distance USE QDS



R.J. Donaldson *et al.*
arXiv:1509.07827 [quant-ph]

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Experimental outlook

- Optimization of experimental parameters to improve the time taken to generate a signature
- Optimization of protocols to improve the time taken to generate a signature
  (Please see Poster Number 53)
- Change of operating parameters to increase transmission range
- Towards lower loss implementation in on-chip systems
- Theoretical model of QDS systems
  (Please see Poster Number 54)

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Advances in Experimental Quantum Digital Signatures

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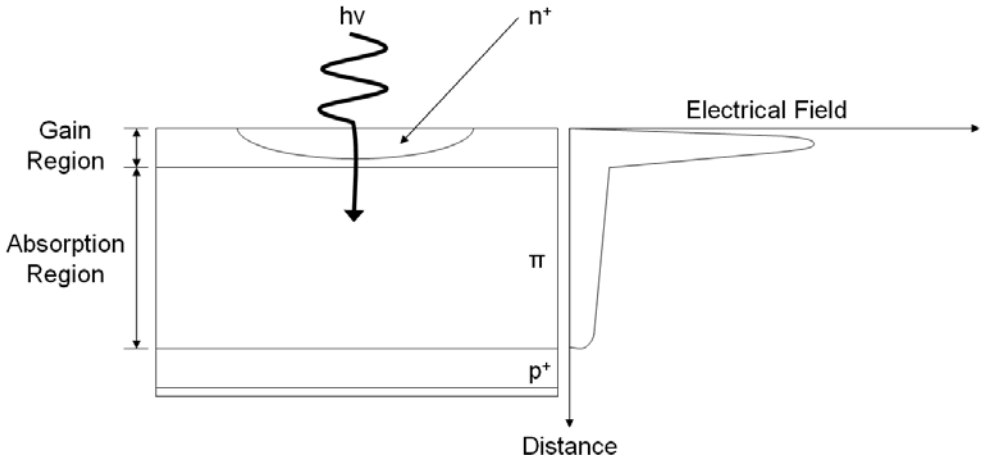
[1]Institute of Photonics & Quantum Sciences, Heriot-Watt University, Edinburgh, United Kingdom
[2]School of Informatics, The University of Edinburgh, Edinburgh, United Kingdom
[3]Institute for Quantum Optics & Quantum Information, Austrian Academy of Sciences, Innsbruck, Austria
[4]Division of Molecular Biology, Ruder Bošković Institute, Zagreb, Croatia
[5]Department of Physics, University of Strathclyde, Glasgow, United Kingdom

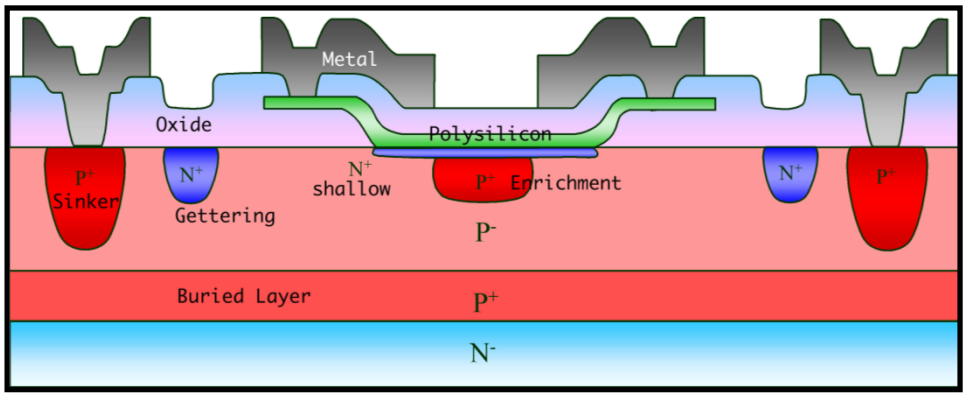
Advances in
Experimental Quantum Digital Signatures

Extra Slides

Single-photon detection

- Semiconductor single-photon avalanche diode (SPAD)
- Detectable current pulse from an electron avalanche
- InGaAs at ~1550 nm wavelength
- InGaAs detectors had relatively high jitter and afterpulsing rates
- Silicon had lower timing jitter and afterpulsing
- Silicon works best in visible and VNIR wavelengths





Detector Cross Sectional Schematic (Silicon)

Superconducting single-photon detectors

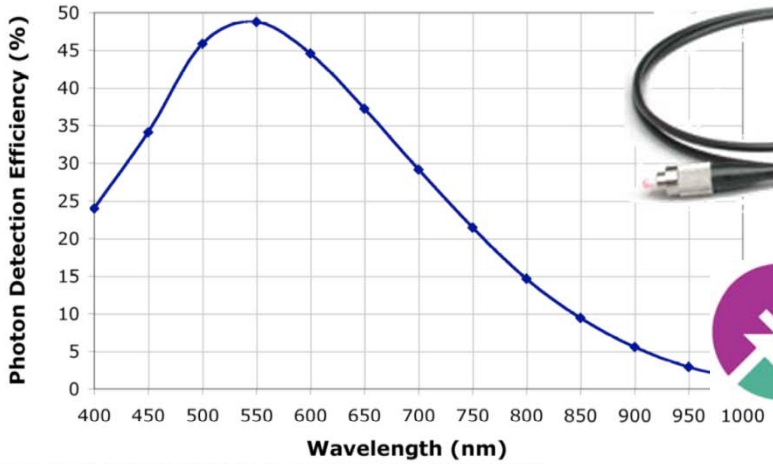


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Semiconductor single-photon detectors

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Timing jitter < 50 ps FWHM





MPD MICRO PHOTON DEVICES

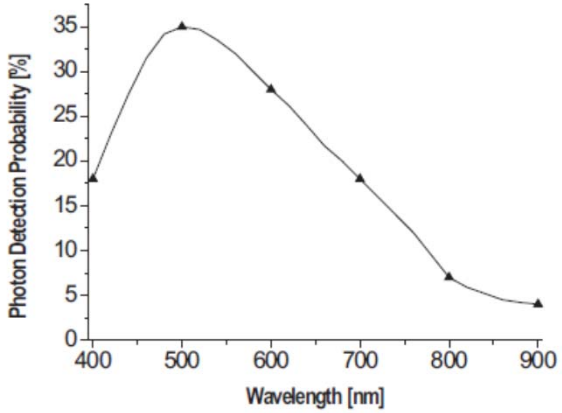
EXCELITAS® TECHNOLOGIES



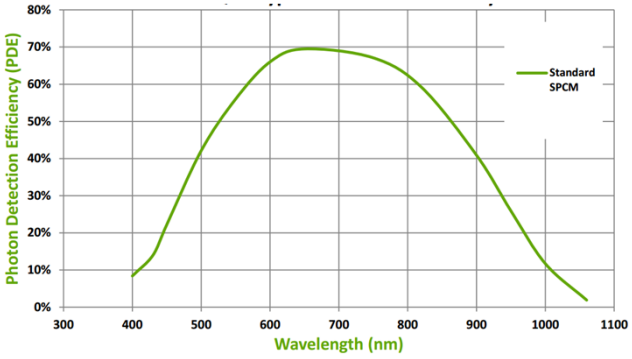
id Quantique



Timing jitter < 60 ps FWHM



Timing jitter < 500 ps FWHM

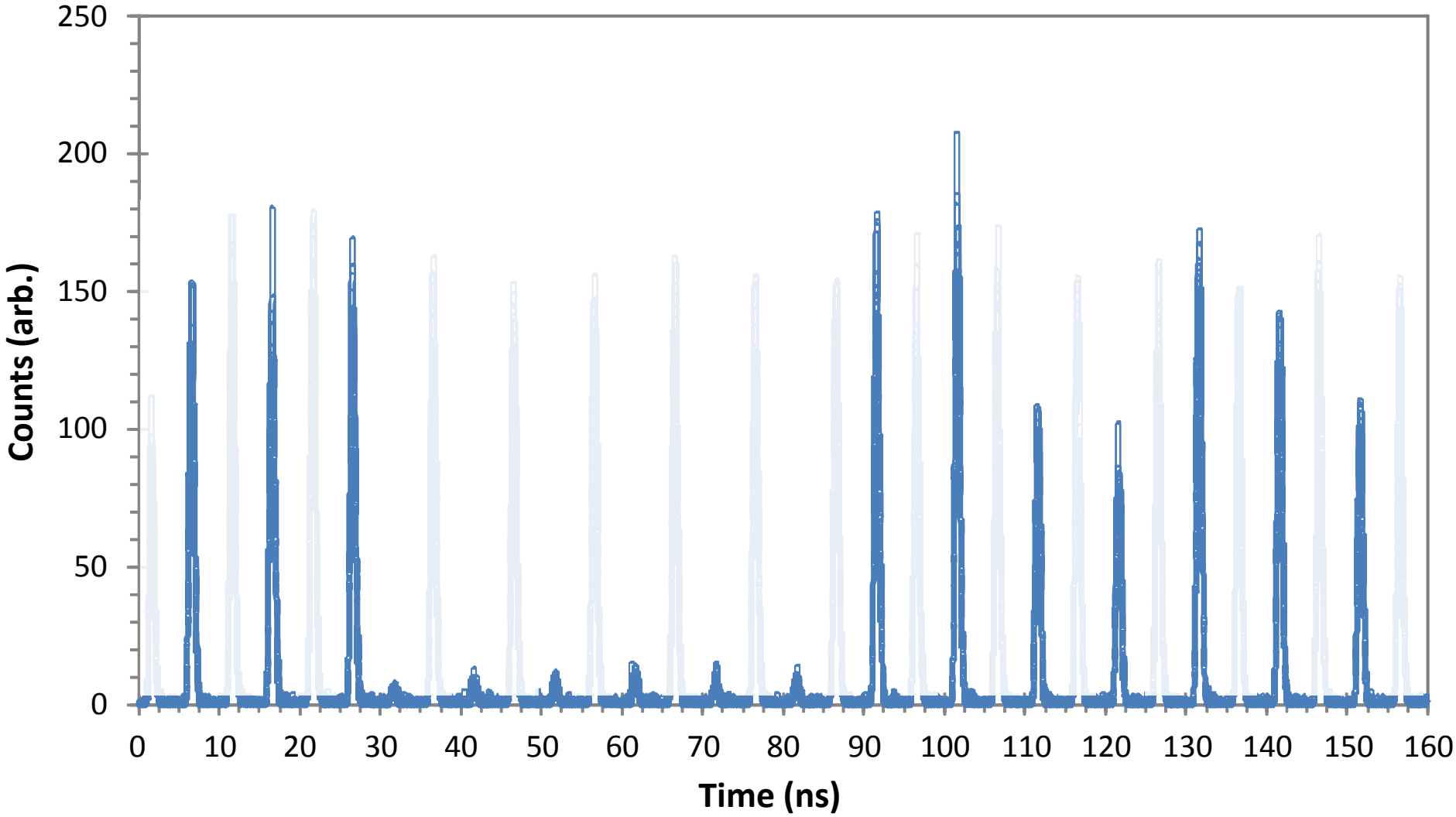


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QDS: Active state discrimination



Nat. Commun. **3** 1174 (2012)

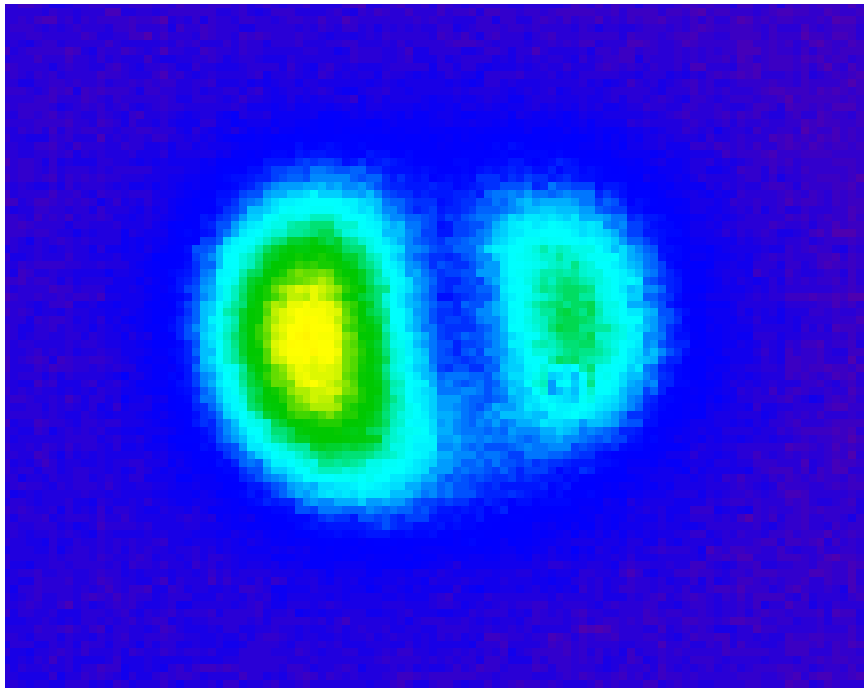
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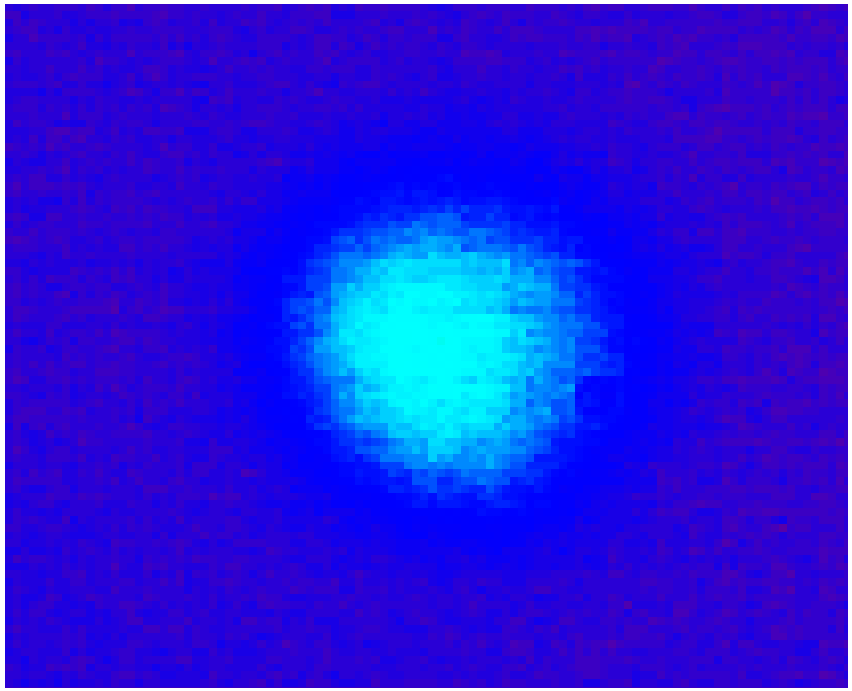
Mode manipulation in “telecomms” fiber

850 nm mode profile in standard telecomms fiber



LP_{01} and LP_{11} modes

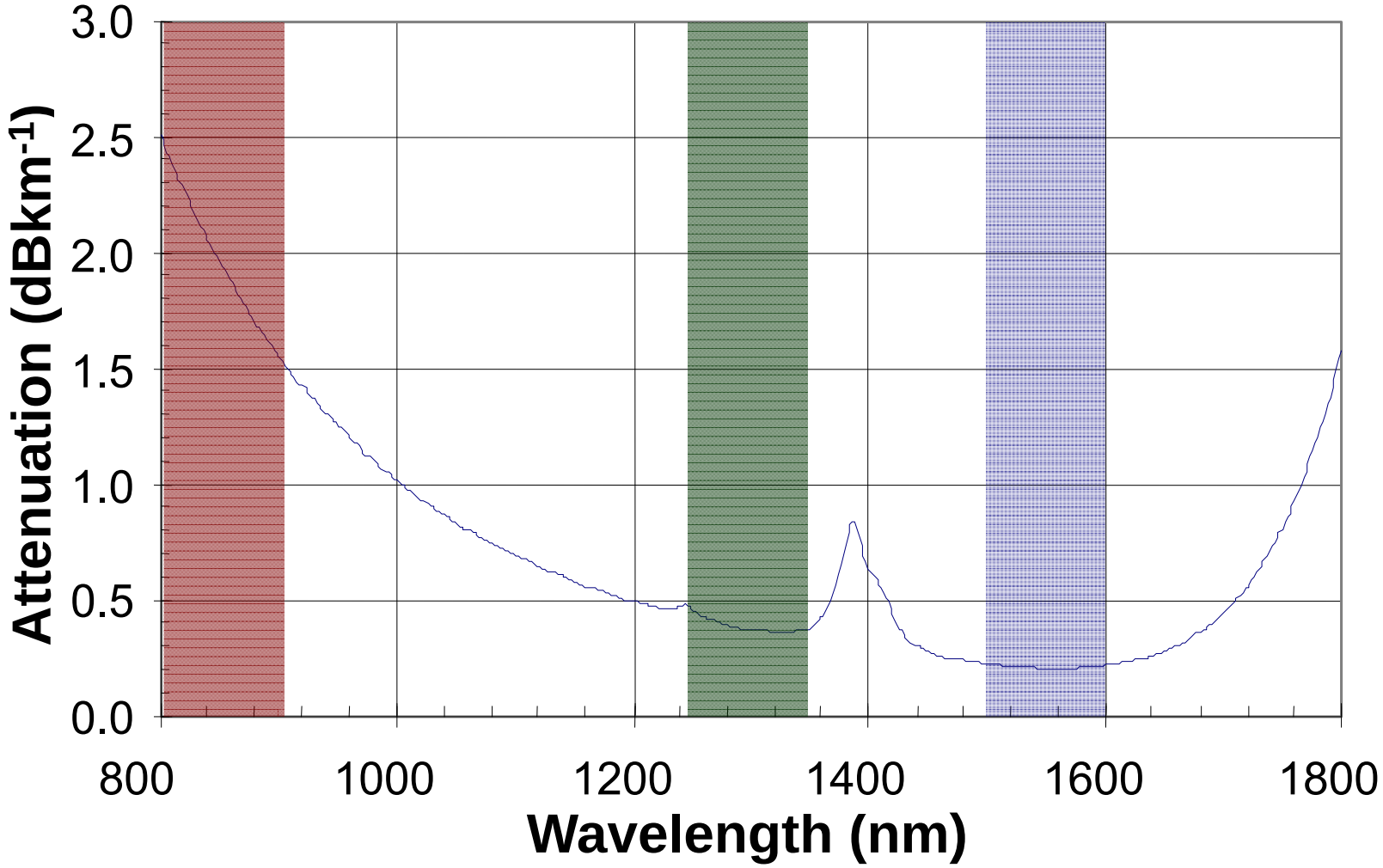
Single mode after mode control



99.6% of photons in LP_{01} mode

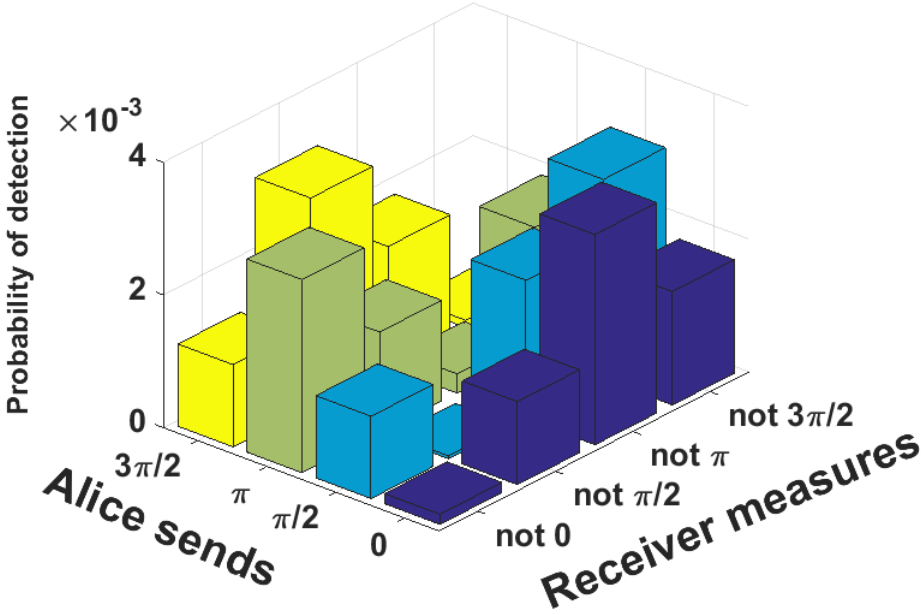
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Choice of operating wavelength



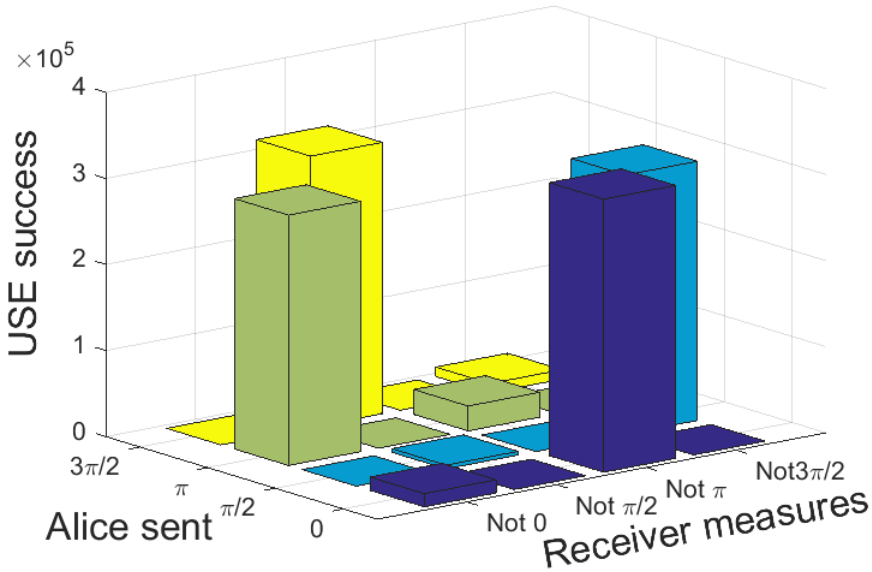
Analysis

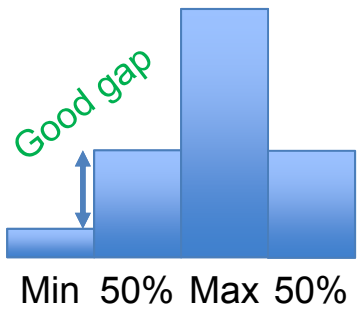
- USE cost matrix.
- Gap.
- Robustness of protocol.
- Probability of forging.
- Effective gap.
- Length of each sequence for each half-bit.
- Authentication/verification thresholds

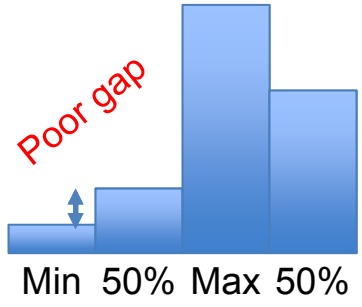


Analysis

- Different from QKD
  - Takes into account all non-orthogonal state measurements.
- Length dependent on:
  - 50% peak ratios
    - Difference in probability between
      - Highest diagonal element
      - Lowest 50% element
  - USE success rate







Improvements over earlier systems

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Protocol	Year	Mean photon number range	Distance range	Gap	Signature length	Time (s)
Multiport and Quantum Memory	2012	0.04-0.28	~5 metres	8.03×10^{-4}	6.91×10^{7}	0.691
Multiport and USD/USE	2014	1-11.5	~5 metres	1.20×10^{-6}	5.13×10^{13}	5.13×10^{5}
Kilometre ranges with USE	2015	0.1-1	~2,000 metres	2.84×10^{-4} (500 m)	1.93×10^{9}	19.3